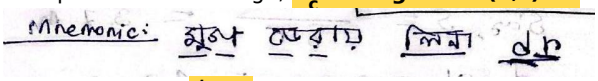
Chapter 8 (V.K. Mehta): All 27 Examples: Quick-Revision Sheet

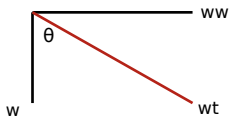
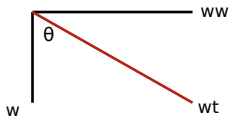
Mechanical Design of Overhead Lines · Power System II · Left: the book's question · Right: your handwritten shortcut solution, digitised

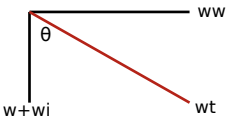
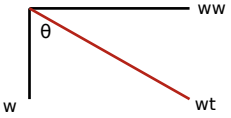
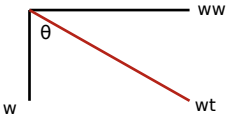
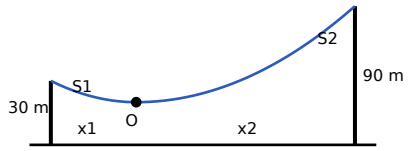
Yellow = formula you wrote · **Gold + bold** = boxed in your notes · **Book answer** = final values from the book (to check your blanks) · Purple = added by Claude where your notes asked (8.15, 8.20, 8.26, 8.27) · ⚠ = a slip in the notes to watch

Problem (V.K. Mehta, Ch-8)	Solution (your handwritten shortcut, digitised)
Type-1: String of 3 (or 4) insulators: $V_2 = V_1(1+K)$, $V_3 = V_1(1+3K+K^2)$	
Example 8.1 (p.171) In a 33 kV overhead line, there are three units in the string of insulators. If the capacitance between each insulator pin and earth is 11% of self-capacitance of each insulator, find (i) the distribution of voltage over 3 insulators and (ii) string efficiency.	**Step-1:** $C_1 = 0.11\ C \Rightarrow K = C_1/C = 0.11$ **Step-2:** $V = 33/\sqrt{3} = 19.05$ kV (string voltage) **Step-3:** $V_2 = V_1(1 + K) =$ ____ $V_3 = V_1(1 + 3K + K^2) =$ ____ **Step-3:** $V = V_1 + V_2 + V_3 = 3.452\ V_1 = 19.05$ $\therefore V_1 =$ ____, $V_2 =$ ____, $V_3 =$ ____ **Step-4:** $\eta = V/(3V_3) \times 100$ **Book answer:** $V_1 = 5.52$ kV, $V_2 = 6.13$ kV, $V_3 = 7.4$ kV; $\eta = 85.8$ %
Example 8.2 (p.172) A 3-phase transmission line is being supported by three disc insulators. The potentials across top unit (i.e., near to the tower) and middle unit are 8 kV and 11 kV respectively. Calculate (i) the ratio of capacitance between pin and earth to the self-capacitance of each unit (ii) the line voltage and (iii) string efficiency.	**Step-1:** $V_1 = 8$ kV, $V_2 = 11$ kV $V_2 = V_1(1 + K) \Rightarrow K =$ ____ **Step-2:** $V_3 =$ ____ $\therefore V = V_1 + V_2 + V_3$ $\therefore$ Line voltage = $\sqrt{3}\ V$, $\eta = V/(3V_3) \times 100$ **Book answer:** $K = 0.375$; $V_3 = 18.12$ kV, $V = 37.12$ kV, line voltage = 64.28 kV; $\eta = 68.28$ %
Example 8.3 (p.172) Each line of a 3-phase system is suspended by a string of 3 similar insulators. If the voltage across the line unit is 17.5 kV, calculate the line to neutral voltage. Assume that the shunt capacitance between each insulator and earth is 1/8th of the capacitance of the insulator itself. Also find the string efficiency.	$K = 1/8$ **Voltage across line unit, $V_3 = 17.5$ kV** $V_2 = V_1(1 + K) = 1.125\ V_1$, $V_3 = V_1(1 + 3K + K^2) = 1.39\ V_1 = 17.5$ $\therefore V_1 =$ ____, $V_2 =$ ____, $V_3 =$ ____ $\therefore V = V_1 + V_2 + V_3 =$ ____ ↳ Line-to-neutral (Line-to-earth) **Book answer:** $V_1 = 12.59$ kV, $V_2 = 14.16$ kV; line-to-neutral V = 44.25 kV; $\eta = 44.25/(3 \times 17.5) \times 100 = 84.28$ %
Example 8.4 (p.173) The three bus-bar conductors in an outdoor substation are supported by units of post type insulators. Each unit consists of a stack of 3 pin type insulators fixed one on the top of the other. The voltage across the lowest insulator is 13.1 kV and that across the next unit is 11 kV. Find the bus-bar voltage of the station.	$V_3 = 13.1$ kV , $V_2 = 11$ kV From this, $K = 0.1$ $\therefore V_1 =$ ____ $\therefore V = V_1 + V_2 + V_3$ (line-to-earth) $\therefore$ bus bar voltage (line) = **$V\sqrt{3}$** **Book answer:** $K = 0.1$ (from $13.1(1+K) = 11(1+3K+K^2)$); $V_1 = 10$ kV; V = 34.1 kV; bus-bar voltage = 59 kV
Example 8.5 (p.173) An insulator string consists of three units, each having a safe working voltage of 15 kV. The ratio of self-capacitance to shunt capacitance of each unit is 8 : 1. Find the maximum safe working voltage of the string. Also find the string efficiency.	$K = 1/8$ safe working voltage = 15 kV $\therefore V_3 = 15$ kV From this $V_2 =$ ____ , $V_1 =$ ____ $\therefore V = V_1 + V_2 + V_3$ and $\eta = V/(3V_3) \times 100$ ↳ voltage across string (safe) **Book answer:** $V_2 = 12.13$ kV, $V_1 = 10.79$ kV; max safe string voltage V = 37.92 kV; $\eta = 84.26$ %
Example 8.8 (p.176) Each conductor of a 3-phase high-voltage transmission line is suspended by a string of 4 suspension type disc insulators. If the potential difference across the second unit from top is 13.2 kV and across the third from top is 18 kV, determine the voltage between conductors.	(add it in Type-1) $V_2 = 13.2$ kV , $V_3 = 18$ kV $\therefore K =$ ____ , $\therefore V_1 =$ ____ $V_4 = V_1(1 + 6K + 5K^2 + K^3)$ $\therefore V = V_1 + V_2 + V_3 + V_4$ ↳ (line-to-earth) $\therefore$ Voltage between conductor = **$\sqrt{3}\ V$** (line voltage) **Book answer:** $K = 0.2$; $V_1 = 11$ kV; $V_4 = 26.49$ kV; V = 68.69 kV; voltage between conductors = 119 kV
Type-2: String of 4 or 5 insulators (let $V_1 = 1$ V, then percentage)	

Problem (V.K. Mehta, Ch-8)	Solution (your handwritten shortcut, digitised)
Example 8.6 (p.174) A string of 4 insulators has a self-capacitance equal to 10 times the pin to earth capacitance. Find (i) the voltage distribution across various units expressed as a percentage of total voltage across the string and (ii) string efficiency.	K = 0.1 Let, $V_1 = 1$ V $\therefore V_2 = V_1 + KV_1 = 1.1$ $V_3 = V_2 + K(V_1 + V_2) = 1.31$ $V_4 = V_3 + K(V_1 + V_2 + V_3) = 1.64$ **percentage voltage distribution:** $\%V_1 = V_1/V \times 100$ = ____ $\%V_2 = V_2/V \times 100$ = ____ $\%V_3 = V_3/V \times 100$ = ____ $\%V_4 = V_4/V \times 100$ = ____ and $\eta = V/(4V_4) \times 100$ ⚠ with K = 0.1 the book gets $V_4 = 1.65$ (V = 5.06); 1.64 belongs to Ex 8.9's V_3 **Book answer:** 19.76 %, 21.74 %, 25.9 %, 32.6 %; η = 76.6 %
Example 8.9 (p.177) A string of four insulators has a self-capacitance equal to 5 times pin to earth capacitance. Find (i) the voltage distribution across various units as a percentage of total voltage across the string and (ii) string efficiency.	Same solution as 8.6 (your notes: "8.6 or 8.9"), with K = 1/5 = 0.2: Let, $V_1 = 1$ V $\therefore V_2 = V_1 + KV_1$ = ____ $V_3 = V_2 + K(V_1 + V_2)$ = ____ $V_4 = V_3 + K(V_1 + V_2 + V_3)$ = ____ **percentage voltage distribution:** $\%V_n = V_n/V \times 100$ and $\eta = V/(4V_4) \times 100$ **Book answer:** $V_2 = 1.2$, $V_3 = 1.64$, $V_4 = 2.408$, V = 6.248 V; 16 %, 19.2 %, 26.3 %, 38.5 %; η = 64.86 %
Example 8.7 (p.175) A string of 5 insulators is connected across a 100 kV line. If the capacitance of each disc to earth is 0.1 of the capacitance of the insulator, calculate (i) the distribution of voltage on the insulator discs and (ii) the string efficiency.	K = 0.1 Let, $V_1 = 1$ V $\therefore V_2 = V_1 + KV_1$ = $V_3 = V_2 + K(V_1 + V_2)$ = $V_4 = V_3 + K(V_1 + V_2 + V_3)$ = **$V_5 = V_4 + K(V_1 + V_2 + V_3 + V_4)$** = $\therefore V = V_1 + V_2 + V_3 + V_4 + V_5$ **percentage: $\%V_1 = V_1/V \times 100$, $\%V_2 = V_2/V \times 100$, $\%V_3 = V_3/V \times 100$, $\%V_4 = V_4/V \times 100$, $\%V_5 = V_5/V \times 100$** Given, voltage across line = 100 kV $\therefore$ string **$V = 100/\sqrt{3}$ kV** **$\therefore$ voltage distribution: $V_1 = \%V_1 \times V$, $V_2 = \%V_2 \times V$, $V_3 = \%V_3 \times V$, $V_4 = \%V_4 \times V$, $V_5 = \%V_5 \times V$** **η:** $\eta = V/(5V_5) \times 100$ **Book answer:** 13.8 %, 15.2 %, 18.2 %, 22.8 %, 30 % of 57.7 kV → $V_1 = 7.96$, $V_2 = 8.77$, $V_3 = 10.5$, $V_4 = 13.16$, $V_5 = 17.3$ kV; η = 66.7 %
Type-3 (less important): Pin-to-line capacitance / guard ring	
Example 8.10 (p.177) The self capacitance of each unit in a string of three suspension insulators is C. The shunting capacitance of the connecting metal work of each insulator to earth is 0.15 C while for line it is 0.1 C. Calculate (i) the voltage across each insulator as a percentage of the line voltage to earth and (ii) string efficiency.	K = C_1/C = 0.15 = pin-to-earth ratio m = 0.1 = pin-to-line ratio Let, $V_1 = 1$ V at top node: **$\therefore V_2 = V_2 + m(V_2 + V_3)$** **$\Rightarrow V_1(1 + K) = V_2 + m(V_2 + V_3)$** and at bottom node: **$V_3 = V_3 + m(V_3)$** **$\Rightarrow V_2 + K(V_1 + V_2) = V_3 + m(V_3)$** From this V_2 = ____ , V_3 = ____ $\therefore V = V_1 + V_2 + V_3$ = ____ **percentage:** $\%V_1 = V_1/V \times 100$, $\%V_2 = V_2/V \times 100$, $\%V_3 = V_3/V \times 100$ and $\eta = V/(3V_3) \times 100$ **Book answer:** $V_2 = 0.943\ V_1$, $V_3 = 1.117\ V_1$ (book: $V_1 = 0.326$V, $V_2 = 0.307$V, $V_3 = 0.364$V); 32.6 %, 30.7 %, 36.4 %; η = 91.5 %

Problem (V.K. Mehta, Ch-8)	Solution (your handwritten shortcut, digitised)
Example 8.11 (p.179) Each line of a 3-phase system is suspended by a string of 3 identical insulators of self-capacitance C farad. The shunt capacitance of connecting metal work of each insulator is 0.2 C to earth and 0.1 C to line. Calculate the string efficiency of the system if a guard ring increases the capacitance to the line of metal work of the lowest insulator to 0.3 C.	$K = C_1/C = 0.2$ $m_1 = 0.1$ (Top) , $m_2 = 0.3$ (bottom) Let, $V_1 = 1$ V **Top: $\therefore V_2 = V_2 + m_1(V_2 + V_3)$** $\Rightarrow$ ____ **bottom: $V_3 = V_3 + m_2(V_3)$** **= ____** $\Rightarrow$ From this V_2 = ____ , V_3 = ____ $\therefore V = V_1 + V_2 + V_3$ $\therefore \eta = V/(3V_3) \times 100$ **Book answer:** top: $V_1(1+K) = V_2 + m_1(V_2+V_3) \rightarrow V_3 = 12V_1 - 11V_2$; bottom: $V_2 + K(V_1+V_2) = V_3 + m_2V_3 \rightarrow 1.3V_3 = 1.2V_2 + 0.2V_1$; $V_2 = 0.993V_1$, $V_3 = 1.077V_1$; $\eta = 95$ %
Type-4: Grading the string (line-to-pin capacitance)	
Example 8.12 (p.179) It is required to grade a string having seven suspension insulators. If the pin to earth capacitance are all equal to C, determine the line to pin capacitance that would give the same voltage across each insulator of the string.	Insulator no = 7 $\therefore$ Node no = 7 − 1 = 6 Line-to-pin capacitance at any node, **$C_n = n/(7 - n) \cdot C$** [n = node no] Node-1: $C_{n1} = 1/(7-1)$ C = (1/6) C Node-2: $C_{n2} = 2/(7-2)$ C , $C_{n3} = 3/(7-3)$ C , $C_{n4} = 4/(7-4)$ C $C_{n5} = 5/(7-5)$ C , $C_{n6} = 6/(7-6)$ C **Book answer:** $C_1 = 0.167C$, $C_2 = 0.4C$, $C_3 = 0.75C$, $C_4 = 1.33C$, $C_5 = 2.5C$, $C_6 = 6C$
Type-5: Corona: critical disruptive voltage	
Example 8.13 (p.184) A 3-phase line has conductors 2 cm in diameter spaced equilaterally 1 m apart. If the dielectric strength of air is 30 kV (max) per cm, find the disruptive critical voltage for the line. Take air density factor $\delta = 0.952$ and irregularity factor $m_0 = 0.9$.	$r = d/2 = 2/2 = 1$ cm spacing, d = 1 m = 100 cm dielectric strength, $g_0 = 30$ kV/cm $= 30/\sqrt{2}$ kV (rms)/cm Given, $m_0 = 0.9$, $\delta = 0.952$ disruptive critical voltage, **$V_c = m_0\, g_0\, \delta\, r \ln(d/r)$ kV**  $\therefore$ Line voltage = **$\sqrt{3}\, V_c$**
Example 8.14 (p.184) A 132 kV line with 1.956 cm dia. conductors is built so that corona takes place if the line voltage exceeds 210 kV (r.m.s.). If the value of potential gradient at which ionisation occurs can be taken as 30 kV per cm, find the spacing between the conductors.	$r = 1.956/2$, $g_0 = 30/\sqrt{2}$, $V_c = 210/\sqrt{3}$ kV Let, $m_0 = 1$, $\delta = 1$ $\therefore V_c = m_0\, g_0\, \delta\, r \ln(d/r)$ kV $\Rightarrow$ d = ____ **Book answer:** $121.25 = 21.2 \times 0.978 \times \ln(d/r) \rightarrow \ln(d/r) = 5.848 \rightarrow d = 348.8\, r = 341$ cm
Example 8.15 (p.185) A 3-phase, 220 kV, 50 Hz transmission line consists of 1.5 cm radius conductor spaced 2 metres apart in equilateral triangular formation. If the temperature is 40°C and atmospheric pressure is 76 cm, calculate the corona loss per km of the line. Take $m_0 = 0.85$.	$V_c = m_0\, g_0\, \delta\, r \ln(d/r)$ kV **[Added by Claude, in your step style; you asked for this one]** $r = 1.5$ cm , $d = 2$ m $= 200$ cm , $m_0 = 0.85$, $g_0 = 21.2$ kV(rms)/cm $\delta = 3.92\, b/(273 + t) = 3.92 \times 76/(273 + 40) = 0.952$ $V_c = 0.85 \times 21.2 \times 0.952 \times 1.5 \times \ln(200/1.5) = 125.9$ kV $V = 220/\sqrt{3} = 127$ kV (phase) Corona loss, **$P = (242.2/\delta)(f + 25)\sqrt{(r/d)}\,(V - V_c)^2 \times 10^{-5}$ kW/km/phase** $= (242.2/0.952)(50 + 25)\sqrt{(1.5/200)}\,(127 - 125.9)^2 \times 10^{-5}$ $= 0.01999$ kW/km/phase $\therefore$ Total loss = 3P = $3 \times 0.01999 = 0.05998$ kW/km **Book answer:** 0.05998 kW per km (all three phases)
Type-6: Corona loss ratio	

Problem (V.K. Mehta, Ch-8)	Solution (your handwritten shortcut, digitised)
Example 8.16 (p.185) A certain 3-phase equilateral transmission line has a total corona loss of 53 kW at 106 kV and a loss of 98 kW at 110.9 kV. What is the disruptive critical voltage? What is the corona loss at 113 kV?	**Formula:** $\mathbf{P \propto (V - V_c)^2}$ Given, $P_1 = 53$ kW , $V_1 = 106/\sqrt{3}$ kV $P_2 = 98$ kW , $V_2 = 110.9/\sqrt{3}$ kV $\therefore P_2/P_1 = (V_2 - V_c)^2 / (V_1 - V_c)^2$ $\Rightarrow \mathbf{V_c =}$ ____ and, $P_3/P_1 = (V_3 - V_c)^2 / (V_1 - V_c)^2 \Rightarrow \mathbf{P_3 =}$ ____ **Book answer:** $V_1 = 61.2$, $V_2 = 64$ kV → $V_c = 54$ kV; $V_3 = 113/\sqrt{3} = 65.2$ kV → $P_3 = (11.2/7.2)^2 \times 53 = 128$ kW
Type-7: Sag (level supports, wind / ice)	
Example 8.17 (p.190) A 132 kV transmission line has the following data: Wt. of conductor = 680 kg/km; Length of span = 260 m; Ultimate strength = 3100 kg; Safety factor = 2. Calculate the height above ground at which the conductor should be supported. Ground clearance required is 10 metres.	w = 680 kg/km = 0.68 kg/m Tension, T = U. strength / Safety factor = 3100/2 = 1550 kg l = 260 m $\therefore$ Sag, $S = wl^2/(8T) = 3.7$ m Ground clearance = 10 m $\therefore$ height = S + 10 = **3.7 + 10** **Book answer:** height = 13.7 m
Example 8.18 (p.190) A transmission line has a span of 150 m between level supports. The conductor has a cross-sectional area of 2 cm². The tension in the conductor is 2000 kg. If the specific gravity of the conductor material is 9.9 gm/cm³ and wind pressure is 1.5 kg/m length, calculate the sag. What is the vertical sag?	l = 150 m , T = 2000 kg wind force, $w_w = 1.5$ kg/m area = 2 cm² **conductor weight/m:** w = Sp. gravity × Volume [m = ρV] = (9.9 g/cm³) × (2 × 1) [V = 2 cm² × 1 m = 2 × 100 cm³] = 1980 g = 1.98 kg **w_t:** $w_t = \sqrt{(w^2 + w_w^2)}$ = $\therefore$ Sag, $S = w_t l^2/(8T)$ = Vertical sag = $S\cos\theta = S \times w/w_t$ **Book answer:** $w_t = 2.48$ kg; S = 3.48 m; θ = 37.23°; vertical sag = 2.77 m 
Example 8.19 (p.191) A transmission line has a span of 200 metres between level supports. The conductor has a cross-sectional area of 1.29 cm², weighs 1170 kg/km and has a breaking stress of 4218 kg/cm². Calculate the sag for a safety factor of 5, allowing a wind pressure of 122 kg per square metre of projected area. What is the vertical sag?	l = 200 m , A = 1.29 cm² w = 1170 kg/km T = Breaking stress × Area / Safety factor = 1088 kg $A = \frac{1}{4}\pi d^2 \Rightarrow d = 1.28$ cm **$\therefore W_w$:** W_w = wind pressure × d $= 122 \times (1.28 \times 10^{-2})$ **W_t:** $W_t = \sqrt{(W^2 + W_w^2)}$ = **S:** $S = W_t l^2/(8T)$ = Vertical sag: $S\cos\theta = S \times W/W_w$ ⚠ $\cos\theta = W/W_t$ (not W/W_w), as in your 8.18 **Book answer:** $w_w = 1.56$ kg; $w_t = 1.95$ kg; S = 8.96 m; θ = 53.13°; vertical sag = 5.37 m 

Problem (V.K. Mehta, Ch-8)	Solution (your handwritten shortcut, digitised)
Example 8.20 (p.191) A transmission line has a span of 275 m between level supports. The conductor has an effective diameter of 1.96 cm and weighs 0.865 kg/m. Its ultimate strength is 8060 kg. If the conductor has ice coating of radial thickness 1.27 cm and is subjected to a wind pressure of 3.9 gm/cm² of projected area, calculate sag for a safety factor of 2. Weight of 1 c.c. of ice is 0.91 gm.	l = 275 m , w = 0.865 kg/m d = 1.96 cm , t = 1.27 cm T = 8060/2 = 4030 kg **[Added by Claude]** $w_i = 0.91 \times \pi t(d + t) \times 100 = 0.91 \times 1288 = 1172$ g = 1.172 kg w_w = pressure × (d + 2t) × 100 = 3.9 × 4.5 × 100 = 1755 g = 1.755 kg $w_t = \sqrt{((w + w_i)^2 + w_w^2)}$ ∴ $S = w_t l^2/(8T)$ **Book answer:** w_t = 2.688 kg; S = 2.688 × 275²/(8 × 4030) = 6.3 m 
Example 8.21 (p.192) A transmission line has a span of 214 metres between level supports. The conductors have a cross-sectional area of 3.225 cm². Calculate the factor of safety under the following conditions: Vertical sag = 2.35 m; Wind pressure = 1.5 kg/m run; Breaking stress = 2540 kg/cm²; Wt. of conductor = 1.125 kg/m run.	l = 214 m , A = 3.225 cm² w = 1.125 kg/m , w_w = 1.5 kg/m ∴ $w_t = \sqrt{(w^2 + w_w^2)}$ ∴ Vertical sag = 2.35 ⇒ S cosθ = 2.35 ⇒ S = 2.35/cosθ = 2.35/(w/w_t) = 3.92 m $S = w_t l^2/(8T)$ ⇒ $T = w_t l^2/(8S)$ = ____ ⇒ B. stress × A / Safety factor = ____ ⇒ Safety factor = ____ **Book answer:** w_t = 1.875 kg; T = 8191/f kg; f = 3 
Example 8.22 (p.192) An overhead line has a span of 150 m between level supports. The conductor has a cross-sectional area of 2 cm². The ultimate strength is 5000 kg/cm² and safety factor is 5. The specific gravity of the material is 8.9 gm/cc. The wind pressure is 1.5 kg/m. Calculate the height of the conductor above the ground level at which it should be supported if a minimum clearance of 7 m is to be left between the ground and the conductor.	l = 150 m , w_w = 1.5 kg/m **w (kg/m):** w = Sp. gravity × volume = (8.9 g/cm³)(2 cm² × 1 m) = 1780 g = 1.78 kg T = (5000/5) × 2 = 2000 kg $w_t = \sqrt{(w^2 + w_w^2)}$ = $S = w_t l^2/(8T)$ = Vertical sag = S cosθ = 2.5 Likewise, ∴ height = vertical sag + clearance = 2.5 + 7 **Book answer:** w_t = 2.33 kg; S = 3.28 m; vertical sag = 2.5 m; height = 9.5 m 
Type-8: Sag with supports at unequal levels: h = $S_2 - S_1$ = (w/2T)($x_2^2 - x_1^2$)	
Example 8.23 (p.193) The towers of height 30 m and 90 m respectively support a transmission line conductor at water crossing. The horizontal distance between the towers is 500 m. If the tension in the conductor is 1600 kg, find the minimum clearance of the conductor and water and clearance mid-way between the supports. Weight of conductor is 1.5 kg/m. Bases of the towers can be considered to be at water level.	 l = 500 = $x_1 + x_2$, w = 1.5 kg/m, T = 1600 kg h = $S_2 - S_1$ ⇒ 90 − 30 = (w/2T)($x_2^2 - x_1^2$) ⇒ $x_2 - x_1$ = ____ ∴ x_1 = ____ , x_2 = ____ ∴ $S_1 = w x_1^2/(2T)$ = 7 m ∴ Min clearance = **$30 - S_1$** = 30 − 7 = 23 **Clearance midway:** x = l/2 − x_1 = 500/2 − 122 = 128 ∴ sag at mid point, $S_{mid} = w x^2/(2T)$ = 7.68 m ∴ Clearance midway = S_{mid} + Min clearance = 7.68 + 23 **Book answer:** $x_2 - x_1$ = 256 m, x_1 = 122 m, x_2 = 378 m; minimum clearance = 23 m; mid-way clearance = 30.68 m

Problem (V.K. Mehta, Ch-8)	Solution (your handwritten shortcut, digitised)
Example 8.24 (p.194) An overhead transmission line conductor having a parabolic configuration weighs 1.925 kg per metre of length. The area of X-section of the conductor is 2.2 cm² and the ultimate strength is 8000 kg/cm². The supports are 600 m apart having 15 m difference of levels. Calculate the sag from the taller of the two supports which must be allowed so that the factor of safety shall be 5. Assume that ice load is 1 kg per metre run and there is no wind pressure.	w = 1.925 kg/m , l = 600 m, h = 15 m w_i = 1 kg/m T = (8000/5) × 2.2 = 3520 kg $w_t = w + w_i$ = 2.925 kg Now, $h = S_2 - S_1$ ⇒ $15 = (w/2T)(x_2^2 - x_1^2)$ ⇒ $x_2 - x_1$ = ____ ∴ x_1 = 270 m , x_2 = 330 m ∴ $S_2 = w_t\, x_2^2/(2T)$ **Book answer:** $x_2 - x_1$ = 60 m; $S_2 = 2.925 \times 330^2/(2 \times 3520)$ = 45.24 m
Example 8.25 (p.195) An overhead transmission line at a river crossing is supported from two towers at heights of 40 m and 90 m above water level, the horizontal distance between the towers being 400 m. If the maximum allowable tension is 2000 kg, find the clearance between the conductor and water at a point mid-way between the towers. Weight of conductor is 1 kg/m.	$h = S_2 - S_1$ ⇒ $90 - 40 = (w/2T)(x_2^2 - x_1^2)$ ⇒ $x_2 - x_1$ = $l = x_2 + x_1$ = 400 m **∴ $x_1 = -50$ m , $x_2 = 450$ m** **clearance mid way:** **S_{mid}:** x = 50 + l/2 x = 50 + 400/2 = 250 ∴ $S_{mid} = w\, x^2/(2T)$ = 15.6 m **S_2:** $S_2 = w\, x_2^2/(2T)$ = 50.6 m ∴ clearance midway = $S_{mid} + (90 - S_2)$ **Book answer:** $x_2 - x_1$ = 500 m; clearance = 15.6 + (90 − 50.6) = 55 m
Example 8.26 (p.195) A transmission line over a hillside where the gradient is 1 : 20, is supported by two 22 m high towers with a distance of 300 m between them. The lowest conductor is fixed 2 m below the top of each tower. Find the clearance of the conductor from the ground. Given that conductor weighs 1 kg/m and the allowable tension is 1500 kg.	**[Added by Claude, in your Type-8 step style; you asked for this one]** Effective tower height = 22 − 2 = 20 m , sinθ = 1/20 w = 1 kg/m , T = 1500 kg , $l = x_1 + x_2$ = 300 m $h = l \sin\theta$ = 300 × 1/20 = 15 m $h = S_2 - S_1$ ⇒ $15 = (w/2T)(x_2^2 - x_1^2)$ ⇒ $x_2 - x_1 = 2Th/(wl)$ = 150 ∴ x_1 = 75 m , x_2 = 225 m ∴ $S_2 = w\, x_2^2/(2T) = 225^2/(2 \times 1500)$ = 16.87 m BC = 20 + 15 = 35 m , GF = $x_1 \tan\theta$ = 75 × 0.05 = 3.75 m ∴ Clearance of lowest point, **OG = BC − S_2 − GF** = 35 − 16.87 − 3.75 **Book answer:** clearance = 14.38 m
Example 8.27 (p.196) A transmission tower on a level ground gives a minimum clearance of 8 metres for its lowest conductor with a sag of 10 m for a span of 300 m. If the same tower is to be used over a slope of 1 in 15, find the minimum ground clearance obtained for the same span, same conductor and same weather conditions.	**[Added by Claude, in your Type-8 step style; you asked for this one]** **Level ground:** $S = wl^2/(8T)$ ⇒ $w/T = 8S/l^2 = 8 \times 10/300^2 = 8/(9 \times 10^3)$ height of tower = Sag + Clearance = 10 + 8 = 18 m **Slope 1 in 15:** h = 300 × 1/15 = 20 m , $x_1 + x_2$ = 300 $h = (w/2T)(x_2^2 - x_1^2)$ ⇒ $x_2 - x_1 = 2Th/(wl)$ = 150 ⇒ x_1 = 75 m, x_2 = 225 m $S_1 = w\, x_1^2/(2T)$ = 2.5 m , $S_2 = w\, x_2^2/(2T)$ = 22.5 m OG = BC − S_2 − GF = (18 + 20) − 22.5 − 5 = 10.5 m [GF = $x_1 \tan\theta$ = 75/15 = 5 m] Clearance at x from O: **$C = x^2/2250 - x/15 + 10.5$** dC/dx = 0 ⇒ 2x/2250 − 1/15 = 0 ⇒ x = 75 m Min clearance = $75^2/2250 - 75/15 + 10.5 = 2.5 - 5 + 10.5$ **Book answer:** minimum ground clearance = 8 m (at 75 m from O, i.e. at the lower tower)